

EduCloud Lite

Build & Deployment Guide

A reproducible guide for the submitted GitHub project

Source: github.com/Funacius/EduCloud

Security note: Replace every placeholder with your own Supabase and AWS values. Never commit passwords, JWT secrets, AWS keys or completed `.env` files.

This guide explains how to build EduCloud Lite from the GitHub repository and run it locally or deploy it to AWS. It intentionally uses placeholders for credentials; each deployment must use its own Supabase project and Cognito User Pool.

Repository: <https://github.com/Funacius/EduCloud>

1. Architecture

Layer	Technology	Purpose
Frontend	React, TypeScript, Vite	Student, Instructor and Admin web UI
Authentication	Amazon Cognito	Signup, email confirmation, login and password recovery
API	FastAPI, SQLAlchemy, Uvicorn	Business rules, authorization and learning workflows
Database	Supabase PostgreSQL	Users, courses, lessons, progress, assessments and certificates
Backend hosting	AWS Elastic Beanstalk	Runs the FastAPI application
Frontend hosting	AWS Amplify Hosting	Builds and serves the Vite production bundle
File delivery	Amazon S3 + CloudFront	Private course assets and cached delivery
Secrets	AWS Systems Manager Parameter Store	Production database URL and JWT secret

2. Prerequisites

- Python 3.11 or newer
- Node.js 18 or newer and npm
- Git
- A Supabase project with its database password
- An Amazon Cognito User Pool and app client in the chosen AWS Region
- AWS CLI configured for the deployment account (only required for AWS deployment)

3. Get the source code

```
git clone https://github.com/Funacius/EduCloud.git
cd EduCloud
```

Use the `main` branch for the submitted version:

```
git checkout main
git pull
```

4. Configure Supabase and Cognito

Create a new Supabase project. In Supabase, open **Connect** → **Connection string** → **Session pooler** and use the URI on port `5432`. Change the scheme to `postgresql+psycopg2://`, URL-encode special characters in the password, and keep `sslmode=require`.

Create a local backend environment file from the template:

```
Copy-Item backend/.env.example backend/.env
```

Edit `backend/.env` with values for your own services:

```
DATABASE_URL=postgresql+psycopg2://postgres.PROJECT_REF:URL_ENCODED_PASSWORD@POOLER_HOST:5432/postgres?sslmode=require
APP_ENV=development
ENABLE_DEV_AUTH=true
JWT_SECRET_KEY=replace-with-a-long-random-value
COGNITO_REGION=YOUR_AWS_REGION
COGNITO_USER_POOL_ID=YOUR_COGNITO_USER_POOL_ID
COGNITO_CLIENT_ID=YOUR_COGNITO_APP_CLIENT_ID
ALLOW_LEGACY_AUTH=true
UPLOAD_STORAGE=local
```

Never commit `backend/.env`, database passwords, Cognito secrets or JWT keys. The file is ignored by Git.

For the frontend, copy the template and use the same Cognito identifiers:

```
Copy-Item frontend/.env.example frontend/.env
```

Set at least:

```
VITE_API_BASE_URL=http://127.0.0.1:8001/api
VITE_COGNITO_REGION=YOUR_AWS_REGION
VITE_COGNITO_USER_POOL_ID=YOUR_COGNITO_USER_POOL_ID
VITE_COGNITO_CLIENT_ID=YOUR_COGNITO_APP_CLIENT_ID
VITE_ALLOW_LEGACY_AUTH=true
```

5. Install and run the backend

```
cd backend
python -m venv .venv
```

```
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements-dev.txt
python -m scripts.check_database
python -c "import main"
python -m scripts.seed_dev_accounts
python -m uvicorn main:app --reload --port 8001
```

The first application start creates the required SQLAlchemy tables in the selected Supabase database. The database check prints connection information without printing the password.

Backend health and API documentation:

- <http://127.0.0.1:8001/health>
- <http://127.0.0.1:8001/docs>

6. Install and run the frontend

Open a second terminal:

```
cd frontend
npm ci
npm run dev
```

Open <http://localhost:5173>. The development seed accounts are:

Role	Email	Password
Student	student@educcloud.local	supplied privately for testing
Instructor	instructor@educcloud.local	supplied privately for testing
Admin	admin@educcloud.local	supplied privately for testing

These accounts are for demonstration only. Do not publish their password in the repository or PDF.

7. Verify the application

Backend tests:

```
cd backend
.\.venv\Scripts\python.exe -m pytest -q
```

Frontend production build:

```
cd frontend
npm run build
npm run preview
```

Recommended smoke test:

1. Register or sign in as a Student and complete the profile.
2. Open a published course, enroll, complete lessons and submit the timed final assessment.
3. Confirm the certificate can be printed or saved as PDF.
4. Sign in as an Instructor, create a course, add lessons and publish a final assessment.

5. Sign in as Admin and verify instructor approval and course oversight.

8. Deploy the backend to Elastic Beanstalk

The backend contains `backend/Procfile`:

```
web: uvicorn main:app --host 0.0.0.0 --port 8000
```

Create an Elastic Beanstalk Python environment in the same AWS Region as Cognito and Parameter Store. Deploy the contents of the `backend` directory. Configure production environment properties, preferably using Systems Manager Parameter Store references:

```
DATABASE_URL      /educcloud/production/database-url
JWT_SECRET_KEY     /educcloud/production/jwt-secret
APP_ENV            production
ENABLE_DEV_AUTH    false
ALLOW_LEGACY_AUTH  false
UPLOAD_STORAGE     s3
AWS_REGION         YOUR_AWS_REGION
COGNITO_REGION     YOUR_AWS_REGION
COGNITO_USER_POOL_ID YOUR_COGNITO_USER_POOL_ID
COGNITO_CLIENT_ID  YOUR_COGNITO_APP_CLIENT_ID
```

The EC2 instance profile must be allowed to read only the two required SSM parameters and to access the private upload bucket. Keep the S3 bucket private.

9. Deploy the frontend to Amplify Hosting

The repository root contains `amplify.yml`:

```
version: 1
applications:
  - appRoot: frontend
    frontend:
      phases:
        preBuild:
          commands:
            - npm ci
        build:
          commands:
            - npm run build
      artifacts:
        baseDirectory: dist
        files:
          - '**/*'
```

Connect the GitHub repository, select the `main` branch, and set the Amplify environment variables:

```
VITE_API_BASE_URL=https://YOUR_CLOUDFRONT_DOMAIN/api
VITE_COGNITO_REGION=YOUR_AWS_REGION
VITE_COGNITO_USER_POOL_ID=YOUR_COGNITO_USER_POOL_ID
VITE_COGNITO_CLIENT_ID=YOUR_COGNITO_APP_CLIENT_ID
VITE_ALLOW_LEGACY_AUTH=false
```

After the build succeeds, open the Amplify URL and test login, course loading, enrollment, assessment and certificate flows.

10. S3 and CloudFront routing

- Keep the S3 upload bucket private.
- Grant CloudFront access with Origin Access Control (OAC), not public-read ACLs.
- Route `/api/*` to Elastic Beanstalk.
- Route `/courses/*` or private asset paths to S3, according to the configured CloudFront behaviors.
- Restrict CORS to the deployed Amplify domain.

11. Production safety checklist

- Replace every `YOUR_*` value before deployment.
- Set `ENABLE_DEV_AUTH=false` and `ALLOW_LEGACY_AUTH=false`.
- Never commit `.env`, passwords, JWT secrets, AWS access keys or Supabase connection strings.
- Enable HTTPS and restrict CORS.
- Review Elastic Beanstalk health and CloudWatch logs after deployment.
- Add AWS Budget alerts and delete unused test resources after evaluation.

12. Troubleshooting

Symptom	Check
<code>sqlalchemy</code> or <code>psycopg2</code> missing	Activate <code>backend/.venv</code> and run <code>pip install -r requirements-dev.txt</code>
Database connection fails	Confirm Session Pooler port <code>5432</code> , URL encoding and <code>sslmode=require</code>
Cognito login fails	Check Region, User Pool ID, app client ID and callback/origin configuration
Amplify shows a blank page	Verify <code>appRoot=frontend</code> , build output <code>dist</code> , and SPA rewrite to <code>/index.html</code>
Elastic Beanstalk is unhealthy	Check <code>Procfile</code> , port <code>8000</code> , environment properties and application logs
Uploads return <code>403</code>	Check the EC2 instance profile, private bucket policy and CloudFront OAC

Submission references

- Source code: <https://github.com/Funacius/EduCloud>
- Workshop/report: <https://funacius.github.io/Fcaj-workshop/>
- Live application: <https://main.djk00b5qbck73.amplifyapp.com/courses>

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